Cartesian Vector of a Line in R²

Recall:

Vector Equation: $\vec{r} = \vec{r}_0 + t\vec{m}$, $t \in \Re$

Vector Equation in Component Form: $(x, y) = (x_0, y_0) + t(a, b)$, $t \in \Re$

Parametric Equations: $\begin{aligned} x &= x_0 + ta \\ y &= y_0 + tb \end{aligned}$, $t \in \Re$

Cartesian Equation of a Line in R²

Determine an equation of a line in **standard form** that contains the point (0 , 5) and has direction vector $\vec{m} = (3,4)$.

← y-int !

↳ $m = \frac{4}{3}$

$\therefore y = \frac{4}{3}x + 5$ ← slope - y-intercept form

$3y = 4x + 15$

$4x - 3y + 15 = 0$ → also called Cartesian Equation

A NORMAL VECTOR is a vector that is PERPENDICULAR to a given vector.

E.g. Determine a normal vector to the vector $\vec{m} = (3,4)$.

Recall if $\vec{m} \perp \vec{n}$, then $\vec{m} \cdot \vec{n} = 0$.

* think neg. reciprocals

shortcut: switch components & change one sign.

$\vec{m} = (3,4)$

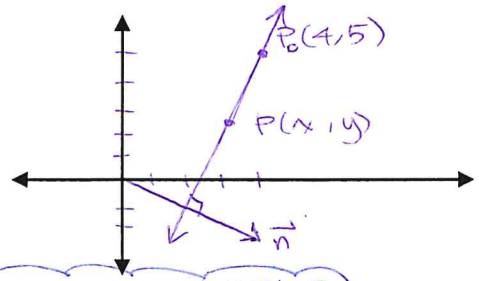
$\vec{n} = (-4,3)$

check: $\vec{m} \cdot \vec{n} = (3,4) \cdot (-4,3)$

$= -12 + 12$

$= 0$ ✓

Example 1: Determine the Cartesian equation of a line whose normal vector is $\vec{n} = (4,-2)$ and the line contains the point (4 , 5).



Let P(x,y) be any point on the line.

$\vec{P_0P} = (x-4, y-5)$ and $\vec{P_0P} \perp \vec{n}$

$\vec{P_0P} \cdot \vec{n} = 0$

$(x-4, y-5) \cdot (4,-2) = 0$

$4x - 16 - 2y + 10 = 0$

$4x - 2y - 6 = 0$ OR $2x - y - 3 = 0$

* Compare normal vector to cartesian equation

Derive using general form: Given point $P_0(x_0, y_0)$ and normal vector $\vec{n} = (A,B)$

Let P(x,y) be any point on the line

$\vec{P_0P} \cdot \vec{n} = 0$

$(x - x_0, y - y_0) \cdot (A,B) = 0$

$Ax - Ax_0 + By - By_0 = 0$

$Ax + By + (-Ax_0 - By_0) = 0$

C.

$\therefore Ax + By + C = 0$

components of normal $\vec{n} = (A,B)$

Cartesian Equation of a Line in R^2

$Ax + By + C = 0$ where $\vec{n} = (A, B)$ is the normal vector.

Example 2: Determine the Cartesian equation of a line through the point (2 , -9) with the normal $\vec{n} = (-4, 5)$.

$Ax + By + C = 0$.

$-4(2) + 5(-9) + C = 0$

$-8 - 45 + C = 0$

$C = 53$

$\therefore -4x + 5y + 53 = 0$

but not start with negative.

$\therefore \boxed{4x - 5y - 53 = 0}$

option #2. use a normal with $A > 0$.

$\vec{n} = (-4, 5)$ but $A > 0$

$\vec{n} = (4, -5)$

$4(2) - 5(-9) + C = 0$

$8 + 45 + C = 0$

$C = -53$

$\therefore \boxed{4x + 5y - 53 = 0}$

Example 3: Given $\vec{m} = (3, 2)$ and a point $P_0(4, -1)$,

a) Determine a vector equation of this line.

$\vec{r} = (4, -1) + t(3, 2)$, $t \in R$

b) Determine the parametric equations of this line.

$x = 4 + 3t$

$y = -1 + 2t$, $t \in R$

c) Determine the Cartesian equation of this line.

$\vec{n} = (2, -3)$

want $A > 0$.

$2x - 3y + C = 0$

$2(4) - 3(-1) + C = 0$

$8 + 3 + C = 0$

$C = -11$

$\therefore \boxed{2x - 3y - 11 = 0}$

Symmetric Equation of a line in R^2

Using the parametric equations, we could create SYMMETRIC EQUATION and use that to create the Cartesian equation.

$x = 4 + 3t$

$x - 4 = 3t$

$\frac{x-4}{3} = t$

$y = -1 + 2t$

$y + 1 = 2t$

$\frac{y+1}{2} = t$

EQUATE!

$\boxed{\frac{x-4}{3} = \frac{y+1}{2}}$ Symmetric Equations

cross-multiply...

$2(x-4) = 3(y+1)$

$2x - 8 = 3y + 3$

$\boxed{2x - 3y - 11 = 0}$